

Tomato — Canonical KB vs. Delta KB (post-fix run)

10 separate diseases (5 + 5, disjoint) | Post-fix half of phase0r_traces (1).jsonl | Tomato | Generated 2026-05-16

Context

After the consolidator token-budget fix the swarm re-ran on Nova. This report uses **only the post-fix half** of the trace file (71 clean Tomato records; the first 147 records are the stale pre-fix run and are excluded). It covers **10 separate Tomato diseases** — 5 in Table A and 5 *different* ones in Table B (the tables are disjoint).

Validation basis. Every `canonical_says` string is itself sourced (Phase-1 KB cites Crop Protection Network / university extension). A delta matching canonical is validated against that source; a delta that diverges is classified in Table B for human adjudication.

Table A — canonical present AND delta has a value (grounded)

5 distinct Tomato diseases. Pre-fix, 92% of deltas had `canonical_says` = "(not specified)"; these show grounding now works.

Disease	Field	Canonical KB said (<code>canonical_says</code>)	Delta KB value (<code>image_shows</code>)	Raw caption (<code>image_quote</code>)
Late Blight	fruit	Dark brown, greasy-to-leathery circular spots.	Tomato fruits with dark brown, greasy-to-leathery circular spots.	The image shows tomatoes with dark brown lesions covering over half of their visible surfaces.
Anthracnose	sporulation	masses of salmon-colored sausage-shaped spores may form on the lesion surface.	The image shows salmon-colored sausage-shaped masses forming on the lesion surface.	The image shows salmon-colored spore masses on the surface of the lesion.
Tomato Spotted Wilt Virus	leaf_necrosis	On tomatoes, uppermost leaves appear bronzed with small brown lesions;	Small brown lesions are scattered across the leaf surface.	The leaf in the image has numerous small brown lesions scattered across its surface.
Phytophthora Blight	concentric_pattern	The image shows a concentric ring lesion on the tomato.	The tomato has a circular brown patch with concentric rings.	The lesion exhibits a concentric structure with alternating light and dark rings.
Southern Blight	look_alikes_visual	Tan to reddish-brown spherical sclerotia (1-2 mm) are visible at the stem base, supporting Southern Blight.	tan to reddish-brown spherical sclerotia (1-2 mm) at the stem base, supporting Southern Blight.	tan to reddish-brown spherical sclerotia (1-2 mm) at the stem base.

Table B — proper changes (delta extends / refines / contradicts canonical)

5 *different* Tomato diseases (no overlap with Table A). The right-hand column classifies the change.

Disease	Canonical KB	Delta KB (image)	The proper change
Tomato Leaf Mould	Pale green or yellowish spots without well defined margins appear on the upper leaf surface and turn a distinctive yellow, with an olive- green mold visible on the lower leaf surface; lesions coalesce when disease is severe, foliage curls, withers, and may drop;	The leaf displays brown spots scattered across its surface, some appearing discrete while others seem to cluster near major veins.	STAGE/COLOR REFINEMENT — brown vein- clustered spots, not canonical’s pale-green/olive mould
Bacterial Speck Of Tomato	Bacterial spot; tomato spotted wilt; bacterial canker of tomato	The image shows a target spot with a central dark brown area surrounded by concentric rings.	ADDED LOOK-ALIKE — describes a target-spot/concentric pattern absent from canonical’s list (flag for review)
Bacterial Canker And Wilt Of Tomato	Dark sunken leaf veins on the tomato plant stems.	Brown streaks are visible in the vascular system through the translucent bark, indicating significant vascular damage.	RELOCATION/REFINEMENT — internal vascular streaking vs canonical’s external sunken veins
Verticillium Wilts	Older/lower leaves yellow (often on one side) with V-shaped areas narrowing from the leaf margin; plants wilt during the hot part of the day but recover in the evening, eventually leaves shrivel, turn brown, and die;	Brown and black necrotic tissue on the leaves, covering large portions of the leaf surface.	STAGE PROGRESSION — extensive brown/black necrosis beyond canonical’s early one-sided V-chlorosis
Early Blight	Lesions first develop on lower leaves as small brownish-black spots that expand to 1/4-1/2 inch with characteristic concentric rings producing a ‘bull’s eye’ or ‘target- spot’ appearance;	The leaf exhibits necrotic tissue in discrete spots, extending inward from veins.	PATTERN REFINEMENT — adds vein-inward necrotic spread alongside canonical’s concentric ‘bullseye’

Verdict

- **Table A** — the fix restored *grounding*: the swarm reads the canonical slice and records confirmations against it across 5 separate Tomato diseases.
- **Table B** — genuine KB *extension*, not captioning, across 5 more Tomato diseases: refinements, relocations, and added differentials.
- Table B deltas are *proposed* refinements for human review before they overwrite canonical — flagged, not yet truth. The Bacterial Speck row in particular (added target-spot pattern) is exactly the kind of item a reviewer should adjudicate.